

Computational Logic

Semantics of propositional logic

Sandra Alves - DCC/FCUP

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Previously on LC...

We have define:

- what constitutes a logic: syntax/language; semantics/meaning; deduction/proofs
- the notion of proposition.
- the syntax/language of propositional logic:
 - propositional variables,
 - connectives and well-formed formulas
 - the notion of sub-formulas.

Semantics of propositional logic

Semantics of propositional logic

Consider the **truth values** **V** and **F** .

Assignment of truth values

$v : \mathcal{V}_{Prop} \longrightarrow \{\mathbf{V}, \mathbf{F}\}$ which assigns a truth value to each propositional

An assignment v can be extended to the set of formulas:

1. for $p \in \mathcal{V}_{Prop}$ $v(p)$ is already set
2. not ϕ :

$$\begin{aligned} v(\neg\phi) &= \mathbf{V} & \text{if } v(\phi) &= \mathbf{F} \\ v(\neg\phi) &= \mathbf{F} & \text{if } v(\phi) &= \mathbf{V} \end{aligned}$$

Semantics of propositional logic

3. ϕ and ψ :

$$\begin{array}{ll} v(\phi \wedge \psi) = \mathbf{V} & \text{if } v(\phi) = \mathbf{V} \text{ and } v(\psi) = \mathbf{V} \\ v(\phi \wedge \psi) = \mathbf{F} & \text{otherwise} \end{array}$$

4. ϕ or ψ :

$$\begin{array}{ll} v(\phi \vee \psi) = \mathbf{V} & \text{if } v(\phi) = \mathbf{V} \text{ or } v(\psi) = \mathbf{V} \\ v(\phi \vee \psi) = \mathbf{F} & \text{otherwise} \end{array}$$

5. if ϕ then ψ :

$$\begin{array}{ll} v(\phi \rightarrow \psi) = \mathbf{F} & \text{if } v(\phi) = \mathbf{V} \text{ and } v(\psi) = \mathbf{F} \\ v(\phi \rightarrow \psi) = \mathbf{V} & \text{otherwise} \end{array}$$

Truth tables

ϕ	$\neg \phi$
F	V
V	F

ϕ	ψ	$\phi \wedge \psi$
F	F	F
F	V	F
V	F	F
V	V	V

ϕ	ψ	$\phi \vee \psi$
F	F	F
F	V	V
V	F	V
V	V	V

ϕ	ψ	$\phi \rightarrow \psi$
F	F	V
F	V	V
V	F	F
V	V	V

If $v(p) = \mathbf{V}$, $v(q) = \mathbf{F}$ and $v(r) = \mathbf{V}$, we can calculate the value of $v((p \wedge q) \vee \neg r)$:

p	q	r	$(p \wedge q)$	$\vee$	$\neg r$
V	F	V	F	F	F

Truth table of a formula

A formula ϕ with n propositional variables has 2^n assignments. Why?

We can build a table in which each row corresponds to one of them:

p	q	r	(p $\wedge$ q)	$\vee$	$\neg$	r
V	V	V	V	V	F	
V	V	F	V	V	V	
V	F	V	F	F	F	
V	F	F	F	V	V	
F	V	V	F	F	F	
F	V	F	F	V	V	
F	F	V	F	F	F	
F	F	F	F	V	V	

Satisfiability, Tautologies and Contradictions

A formula ϕ is:

satisfiable if there is an assignment v such that $v(\phi) = \mathbf{V}$,
write $\models_v \phi$ and say that v satisfies ϕ

tautology if for all assignments v , $v(\phi) = \mathbf{V}$ and write $\models \phi$
Ex: $\models p \vee \neg p$ (Third excluded)

contradiction if for all assignment v , $v(\phi) = \mathbf{F}$ and write $\not\models \phi$.
Ex: $\not\models p \wedge \neg p$.

A formula is **unsatisfiable** if and only if it is a contradiction.

Satisfiability relation

Let v be an assignment, the relation $\models_v$ can be defined inductively in the structure of ϕ by:

1. $\models_v p$ iff $v(p) = \mathbf{V}$;
2. $\models_v \neg\phi$ iff $\not\models_v \phi$;
3. $\models_v \phi \wedge \psi$ iff $\models_v \phi$ and $\models_v \psi$;
4. $\models_v \phi \vee \psi$ iff $\models_v \phi$ or $\models_v \psi$;
5. $\models_v \phi \rightarrow \psi$ iff $\not\models_v \phi$ or $\models_v \psi$;

We say that v *satisfies* ϕ if $\models_v \phi$.

Example

Let $v(p) = \mathbf{V}$, $v(q) = v(r) = \mathbf{F}$ determine whether

$$\models_v (p \rightarrow (q \vee r)) \vee (r \rightarrow \neg p).$$

To do so, $\models_v (p \rightarrow (q \vee r))$ or $\models_v (r \rightarrow \neg p)$.

The second is verified because $\not\models_v r$.

So we can also conclude that

$$\models_v (p \rightarrow (q \vee r)) \vee (r \rightarrow \neg p).$$

Example

Determine whether $(p \wedge q) \vee \neg r$ is valid (tautology), that is, $\models (p \wedge q) \vee \neg r$.

But, if $v(r) = \mathbf{V}$ and $v(p) = \mathbf{F}$ then

$$\not\models_v (p \wedge q) \vee \neg r$$

because

$$\not\models_v \neg r \text{ and } \not\models_v (p \wedge q).$$

So, it's not valid.

Satisfiability of a set of formulas

Let Γ be a set of formulas.

An assignment v **satisfies** Γ (notation $\models_v \Gamma$) if and only if v satisfies every formula $\psi \in \Gamma$.

$$\forall \psi \in \Gamma, \models_v \psi$$

Γ is **satisfiable** if there is an assignment that satisfies it.

Example

Let $\Gamma = \{p \vee q, \neg q\}$:

- for $v_1(p) = \mathbf{F}$ and $v_1(q) = \mathbf{V}$, $\not\models_{v_1} \Gamma$, since $\not\models_{v_1} \neg q$;
- but for $v_2(p) = \mathbf{V}$ and $v_2(q) = \mathbf{F}$, Γ is satisfiable.

Semantic consequence and equivalence

A formula ϕ is a **semantic consequence** of Γ if every assignment v that satisfies Γ also satisfies ϕ (that is, $v(\phi) = \mathbf{V}$). We denote this by $\Gamma \models \phi$.

Notation:

- $\emptyset \models \phi$ is **equivalent** to $\models \phi$
- If $\Gamma = \{\psi\}$ and $\Gamma \models \phi$ then ϕ is said to be a **semantic consequence** of ψ ($\psi \models \phi$)

If $\psi \models \phi$ and $\phi \models \psi$ then ψ and ϕ are **semantically equivalent** and we write $\psi \Leftrightarrow \phi$.

Semantically equivalent formulas

$\phi \wedge \psi$	$\Leftrightarrow$	$\psi \wedge \phi$	(commutativity)
$\phi \vee \psi$	$\Leftrightarrow$	$\psi \vee \phi$	(commutativity)
$\neg(\phi \wedge \psi)$	$\Leftrightarrow$	$(\neg\phi \vee \neg\psi)$	(DeMorgan's Law)
$\neg(\phi \vee \psi)$	$\Leftrightarrow$	$(\neg\phi \wedge \neg\psi)$	(DeMorgan's Law)
$(\phi \wedge \psi) \wedge \theta$	$\Leftrightarrow$	$\phi \wedge (\psi \wedge \theta)$	(associativity)
$(\phi \vee \psi) \vee \theta$	$\Leftrightarrow$	$\phi \vee (\psi \vee \theta)$	(associativity)
$(\phi \vee \phi)$	$\Leftrightarrow$	ϕ	(idempotence)
$(\phi \wedge \phi)$	$\Leftrightarrow$	ϕ	(idempotence)
$(\phi \wedge \psi) \vee \theta$	$\Leftrightarrow$	$(\phi \vee \theta) \wedge (\psi \vee \theta)$	(distributiveness)
$(\phi \vee \psi) \wedge \theta$	$\Leftrightarrow$	$(\phi \wedge \theta) \vee (\psi \wedge \theta)$	(distributivity)
$\neg\neg\phi$	$\Leftrightarrow$	ϕ	(Double negation)
$\phi \rightarrow \psi$	$\Leftrightarrow$	$\neg\phi \vee \psi$	

Complete sets of connectives

We can restrict ourselves to only complete sets of connectives.

Thus, we could have considered in the definition of the language of propositional logic, only:

- the connectives $\wedge$, $\vee$ and $\neg$
- the connectives $\rightarrow$ and $\neg$
- ...

And treat the rest as abbreviations.

One of the **advantages** would be to have a smaller number of types of formulas, which is good for proofs...

More connectives

But we can also define other connectives. For example, one for each of the unary or binary truth functions... The most common are:

Designation	Connective	Semantically equivalent formula
False	F	$\phi \wedge \neg\phi$
Truth	V	$\phi \vee \neg\phi$
Implication	$\phi \rightarrow \psi$	$\neg\phi \vee \psi$
Equivalence	$\phi \leftrightarrow \psi$	$(\neg\phi \vee \psi) \wedge (\phi \vee \neg\psi)$
Exclusive Or	$\phi \dot{\vee} \psi$	$(\phi \wedge \neg\psi) \vee (\neg\phi \wedge \psi)$
Not-and	$\phi \tilde{\wedge} \psi$	$\neg(\phi \wedge \psi)$
Not-or	$\phi \tilde{\vee} \psi$	$\neg(\phi \vee \psi)$

Exercise

Build the truth tables associated with each of the connectives.

Truth (or Boolean) functions

p	q	$p \wedge q$
F	F	F
F	V	F
V	F	F
V	V	V

x	y	$F(x, y)$
F	F	F
F	V	F
V	F	F
V	V	V

The truth table of a formula ϕ , with $n > 0$ propositional variables $p_1, \dots, p_n$, defines a truth function

$$F_\phi : \{\mathbf{V}, \mathbf{F}\}^n \longrightarrow \{\mathbf{V}, \mathbf{F}\}$$

such that $F_\phi(x_1, \dots, x_n) = v_X(\phi)$, where $v_X(p_i) = x_i$ for $i \in \{1 \dots n\}$ and $x_i \in \{\mathbf{V}, \mathbf{F}\}$

Truth Functions

Any function of $f : \{\mathbf{V}, \mathbf{F}\}^n \longrightarrow \{\mathbf{V}, \mathbf{F}\}$, with $n > 0$ is called a truth function.

For $n = 1$, how many functions do we have?

x_1	$f_1(x_1)$	x_1	$f_2(x_1)$	x_1	$f_3(x_1)$	x_1	$f_4(x_1)$
V	F	V	F	V	V	V	V
F	F	F	V	F	F	F	V

There are 4 truth functions of arity 1. How about arity 2?

There are 16 truth functions of arity 2. Try it!

Allennd how many functions exist of arity n , for $n > 0$?

Complete set of connectives

A set of connectives C is said to be **complete** if for any truth function f there is a formula ϕ with n propositional variables and containing only connectives of C , such that $F_\phi = f$

Proposition

The connective set $\{ \wedge, \vee, \neg \}$ is complete.

Proof: We prove by induction on n .

- **Base.** For $n = 1$ consider the 4 truth functions:

x_1	f_1	x_1	f_2	x_1	f_3	x_1	f_4
V	F	V	F	V	V	V	V
F	F	F	V	F	F	F	V

Where $\phi_1 = p \wedge \neg p$, $\phi_2 = \neg p$, $\phi_3 = p$, $\phi_4 = p \vee \neg p$, we have that $F_{\phi_i} = f_i$ for $1 \leq i \leq 4$.

Complete set of connectives

Proof (cont.):

- **Induction.** Assuming that the hypothesis is valid for n , let $f : \{\mathbf{V}, \mathbf{F}\}^{n+1} \rightarrow \{\mathbf{V}, \mathbf{F}\}$ be given. We construct two n -ary functions f_1 and f_2 such that:

$$f_1(x_1, \dots, x_n) = f(x_1, \dots, x_n, \mathbf{V})$$

$$f_2(x_1, \dots, x_n) = f(x_1, \dots, x_n, \mathbf{F}).$$

By induction hypothesis there exist ϕ_i , with variables $p_1, \dots, p_n$, such that $F_{\phi_i} = f_i$ for $i = 1, 2$. Take $\phi = (p_{n+1} \wedge \phi_1) \vee (\neg p_{n+1} \wedge \phi_2)$, then $F_\phi = f$.

Proposition

The $\{\neg, \rightarrow\}$ connective set is complete.

Proof: Note that $\phi \wedge \psi \Leftrightarrow \neg(\phi \rightarrow \neg\psi)$ and $\phi \vee \psi \Leftrightarrow (\neg\phi \rightarrow \psi)$.